

Fruit, vegetable, and fish consumption and heart rate variability: the Veterans Administration Normative Aging Study

Background: Higher intakes of fruit, vegetables, and dark fish may prevent sudden cardiac death and arrhythmias, but the exact mechanisms are not fully understood. Objective: We examined whether high consumption of fruit, vegetables, and dark fish would be associated with beneficial changes in heart rate variability (HRV). Design: HRV variables were measured among 586 older men with 928 total observations from November 2000 to June 2007 in the Normative Aging Study, a community-based longitudinal study of aging. Dietary intake was evaluated with a self-administered semiquantitative food-frequency questionnaire and categorized into quartiles. Results: After controlling for potential confounders, intake of green leafy vegetables was positively associated with normalized high-frequency power and inversely associated with normalized low-frequency power (P for trend < 0.05). These significant associations were retained after further adjustment for healthy lifestyle factors, such as physical activity and use of multivitamins. No significant association was seen between HRV measures and intakes of other fruit and vegetables, vitamin C, carotenoids, tuna and dark-meat fish, or $n-3$ (omega-3) fatty acids. An effect modification of intake of noncitrus fruit by obesity and of total vegetables and cruciferous vegetables by cigarette smoking was seen, which warrants further investigation. Conclusion: These findings suggest that higher intake of green leafy vegetables may reduce the risk of cardiovascular disease through favorable changes in cardiac autonomic function.